Cells have rectangular oblique or tapering ends and persistent protoplast. They possess thick primary non-lignified walls.They provide mechanical support for growing organs.

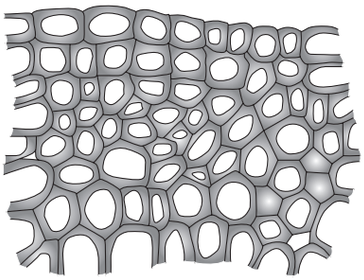


Figure 18.3 Collenchyma

Sclerenchyma: Sclerenchyma consists of thick walled cells which are often lignified. Sclerenchyma cells are dead and do not possess living protoplasts at maturity. Sclerenchyma cells are grouped into fibres and sclereids.

Fibres are elongated sclerenchymatous cells, usually with pointed ends. Their walls are lignified. Fibres are abundantly found in many plants. The average length of fibres is 1 to 3 mm, however in plants like *Linum usitatissimum* (flax), *Cannabis sativa* (hemp) and *Corchorus capsularis* (jute), fibres are extensively longer, ranging from 20 mm to 550 mm.

Sclereids are widely distributed in plant body. They are usually broad, may occur in single or in groups. Sclereids are isodiametric, with liginified walls. Pits are prominent and seen along the walls. Lumen is filled with wall materials. Sclereids are also common in fruits and seeds.

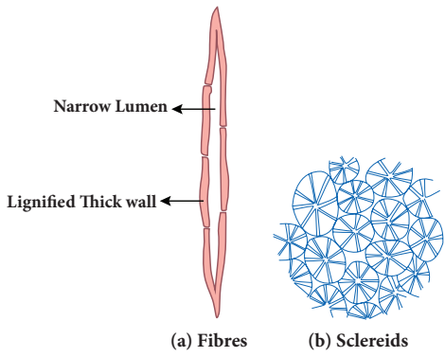


Figure 18.4 Sclerenchyma (a) Fibres, (b) Sclereids

B. Complex tissues

Complex tissues are made of more than one type of cells that work together as a unit. Complex tissues consist of parenchyma and sclerenchyma cells. However, collenchymatous cells are not present in such tissues. Common examples are xylem and phloem.

i. Xylem

Xylem is a conducting tissue which conducts water, mineral nutrients upward from root to leaves. Xylem gives mechanical support to the plant body. Xylem is composed of: (i) xylem tracheids (ii) xylem fibres (iii) xylem vessels and (iv) xylem parenchyma.

Xylem tracheids: These are elongated or tube-like dead cells with hard, thick and

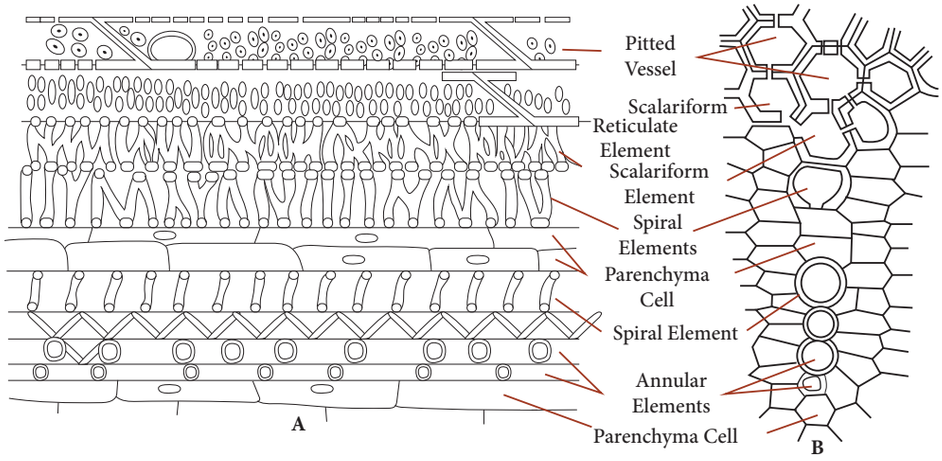


Figure 18.5 A. xylem longitudinal section B. xylem transverse section

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